Large-Mass Expansion and Limit-Cycle Correction in the Underdamped RTP-Spring Model

1 Effective Dynamics

We consider the underdamped spring-coupled dynamics of the probe position $X(t)$, spring center $X_s(t) = v_0 t$, and probe velocity $v(t) = \dot{X}(t)$. Let

$$x(t) = X(t) - X_s(t), \qquad u(t) = v(t) - v_0.$$

The mean-field equation used to interpret the simulations is

$$\dot{x} = v - v_0 = u, \tag{1}$$
$$M\dot{v} = f(v) - kx. \tag{2}$$

Here k is the spring constant and $f(v)$ is the active force-velocity curve. In the current simulations the per-particle force is rescaled so that the total mean-field force is approximately independent of N. Thus the same mean-field curve $f(v)$ should be used for different N.

The fixed point associated with imposed velocity v_0 is

$$v_* = v_0, \qquad x_* = \frac{f(v_0)}{k}.$$

Writing

$$y = x - x_*, \qquad u = v - v_0,$$

we obtain

$$\dot{y} = u, \tag{3}$$
$$M\dot{u} = f(v_0 + u) - f(v_0) - ky. \tag{4}$$

2 Large-M Expansion

Let

$$\epsilon = M^{-1/2}, \qquad \tau = \epsilon t.$$

Then

$$\frac{d}{dt} = \epsilon \frac{d}{d\tau}.$$

The equations become

$$\frac{dy}{d\tau} = u, \tag{5}$$
$$\frac{du}{d\tau} = -ky + \epsilon\left[f(v_0 + u) - f(v_0)\right]. \tag{6}$$

Equivalently,

$$\frac{d^2y}{d\tau^2} + ky = \epsilon\left[f\left(v_0 + \frac{dy}{d\tau}\right) - f(v_0)\right].$$

Therefore, at leading order in $M^{-1/2}$, the motion is harmonic:

$$u(\tau) \simeq A\cos\Big(\sqrt{k}\tau + \phi\Big), \tag{7}$$

$$y(\tau) \simeq \frac{A}{\sqrt{k}}\sin\Big(\sqrt{k}\tau + \phi\Big). \tag{8}$$

In the original time variable,

$$v(t) \simeq v_0 + A(t)\cos\left(\sqrt{\frac{k}{M}}t + \phi_0\right), \tag{9}$$

$$x(t) \simeq x_* + \frac{A(t)}{\sqrt{k}}\sin\left(\sqrt{\frac{k}{M}}t + \phi_0\right). \tag{10}$$

The leading oscillation period is

$$T \simeq 2\pi\sqrt{\frac{M}{k}}.$$

3 Amplitude Equation for a Cubic Normal Form

Near the unstable part of the mean-field curve, a useful local normal form is an odd cubic force law

$$f_{\text{cub}}(v) = c_1 v + c_3 v^3, \qquad c_3 < 0.$$

For the idealized force

$$f(v) = 300v - 10v^3,$$

we have $c_1 = 300$ and $c_3 = -10$.

Averaging over the fast harmonic oscillation gives the slow amplitude equation

$$\frac{dA}{dt} = \frac{A}{M}\left[\frac{1}{2}f'_{\text{cub}}(v_0) + \frac{1}{16}f'''_{\text{cub}}A^2\right].$$

Since

$$f'_{\text{cub}}(v_0) = c_1 + 3c_3v_0^2, \qquad f'''_{\text{cub}} = 6c_3,$$

the nonzero stable amplitude satisfies

$$A_{\text{LC}}^2 = -\frac{8f'_{\text{cub}}(v_0)}{f'''_{\text{cub}}} = -\frac{4\left(c_1 + 3c_3v_0^2\right)}{3c_3},$$

provided $f'_{\text{cub}}(v_0) > 0$.

For the idealized force $300v - 10v^3$,

$$f'(v_0) = 300 - 30v_0^2 = 30(10 - v_0^2),$$

and therefore

$$A_{\text{LC}} = 2\sqrt{10 - v_0^2}, \qquad |v_0| < \sqrt{10}.$$

Thus the large-M limit-cycle approximation is

$$v(t) \simeq v_0 + 2\sqrt{10 - v_0^2}\cos\left(\sqrt{\frac{k}{M}}t + \phi_0\right), \tag{11}$$

$$x(t) \simeq \frac{f(v_0)}{k} + \frac{2\sqrt{10 - v_0^2}}{\sqrt{k}}\sin\left(\sqrt{\frac{k}{M}}t + \phi_0\right). \tag{12}$$

4 Energy-Normalized Radius

The natural phase-plane radius follows from the oscillator energy

$$E = \frac{1}{2}Mu^2 + \frac{1}{2}ky^2.$$

The two terms are comparable when

$$u^2 \sim \frac{k}{M}y^2.$$

Therefore the displacement coordinate should be normalized as

$$q = \sqrt{\frac{k}{M}}\,y.$$

Equivalently, in terms of spring force kx,

$$q = \frac{k(x - x_*)}{\sqrt{Mk}} = \frac{kx - f(v_0)}{\sqrt{Mk}}.$$

The predicted limit cycle is a circle in the (q, u) plane:

$$q^2 + u^2 \simeq A_{\mathrm{LC}}^2.$$

This is why the trajectory plots use the x-direction rescaling by $\sqrt{M}$, and why the measured radius is computed as

$$r(t) = \sqrt{\left[\frac{kx(t) - f(v_0)}{\sqrt{Mk}}\right]^2 + [v(t) - v_0]^2}.$$

5 Correction to the Mean Spring Force

The observed average spring force is not simply $f(v_0)$ once the trajectory has a finite velocity amplitude. Since the probe samples values

$$v(t) = v_0 + u(t),$$

the relevant mean force is

$$\langle f(v_0 + u)\rangle.$$

Expanding around v_0,

$$\langle f(v_0 + u)\rangle = f(v_0) + \frac{1}{2}f''(v_0)\langle u^2\rangle + \frac{1}{24}f^{(4)}(v_0)\langle u^4\rangle + \cdots.$$

For an approximately sinusoidal limit cycle

$$u(t) \simeq A\cos\theta, \qquad \langle u^2 \rangle = \frac{A^2}{2}.$$

For the cubic normal form, $f^{(4)} = 0$, so the leading center shift is

$$\Delta f_{\text{center}} = \frac{1}{4} f''_{\text{cub}}(v_0) A^2_{\text{LC}}.$$

Since

$$f''_{\text{cub}}(v_0) = 6c_3 v_0, \qquad A^2_{\text{LC}} = -\frac{4f'_{\text{cub}}(v_0)}{3c_3},$$

we get the compact expression

$$\boxed{\Delta f_{\text{center}} = -2v_0 f'_{\text{cub}}(v_0)}.$$

For the idealized force $300v - 10v^3$,

$$\Delta f_{\text{center}} = -60v_0(10 - v_0^2), \qquad |v_0| < \sqrt{10}.$$

For $v_0 > 0$, this correction is negative, explaining the observed shift of the cycle center toward negative spring force at finite M.

6 Connection to the Plots

In the simulations with $k = k_s = 250$, the steady-state comparison plot uses

$$y_{\text{plot}} = k_s dX \times \texttt{force\_scale}, \qquad \texttt{force\_scale} = \frac{120}{k_s}.$$

The plotted mean-field baseline uses the measured mean-field data directly through interpolation. The limit-cycle correction is computed from the original local odd-cubic normal form at each v_0,

$$f_{\text{cub}}(v) = c_1 v + c_3 v^3,$$

where c_1, c_3 are fitted from the odd part of the mean-field data. The corrected prediction is

$$f_{\text{MF}}(v_0) + \Delta f_{\text{center}}(v_0),$$

with

$$\Delta f_{\text{center}}(v_0) = -2v_0 f'_{\text{cub}}(v_0).$$

The correction is applied only where $f'_{\text{cub}}(v_0) > 0$, i.e. where the fixed point is predicted to be weakly unstable by the local cubic model.